

Kutay Kulbak

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Summary

Research oriented undergraduate student majoring in physics and mathematics with experience in nonlinear dynamics, spintronics, data analysis, and Olympiad level physics education. Interested in theoretical physics and numerical models, and eager to join research groups working on fundamentals of physics.

Education

École Polytechnique - Bachelor of Science

Sept. 2024 – June 2027

Palaiseau, France

BS in Physics & Mathematics (Cumulative GPA: 4.04 out of 4.3)

Awarded Excellence Scholarship for high academic performance.

Bahçeşehir High School for Science and Technology

Sept. 2019 – June 2024

Istanbul, Turkey

Research Experience

Research at Irradiated Solids Laboratory (LSI) - École Polytechnique

Since December 2024

Undergraduate Research Assistant - Supervised by Prof. Vasily Temnov

- Analyzed the Gilbert damping behavior of small amplitude magnetization precessions governed by Landau-Lifshitz-Gilbert equation in anisotropic nanomagnets using theoretical and numerical methods.
- Derived exact expressions relating precession parameters to local properties of the free energy at stable equilibrium positions.
- Identified the bifurcation region where the usual assumption on the quality factor of the precession fails.
- Presented poster on the research results at SPIE Photonics Europe 2026 conference on April 15, 2026. Received Student Travel Grant to present the poster in person. [Presentation Information](#)
- Currently working on further analysis of the damping phenomenon, including implementation of the results to time-dependent configurations and magnetization switching analyses.
- See the manuscript on <https://arxiv.org/abs/2604.16938>.

Research at Leprince Ringuet Laboratory (LLR) - École Polytechnique

Since December 2025

Laboratory Research Project - Coursework

- Studied the variable distributions in SMOG2 (Fixed-Target) collision simulations for the LHCb experiment of CERN using ROOT software, with a focus on χ_c productions in pAr, ppAr, pHe, ppHe and PbAr collisions.
- Wrote a project report summarizing the validation results and gave an oral defense.

Academic Activities & Experience

PLAS@PAR Summer School

August 2025

Participant

- Participated in lectures, presentations and experimental demonstrations on foundations and applications of plasma physics in the one week summer school.
- Worked on numerical simulations of simple particle dynamics in external magnetic fields.

Private Physics Tutoring

Sept. 2024 – Feb. 2025

Tutor

- Tutored and mentored a student preparing for Turkish national physics Olympiads.
- Curated and solved problem sets with selected classical Olympiad problems and questions of my own.
- Mentee won the bronze medal in the 2024 National Physics Olympiad.

Physics Olympiads

Sept. 2019 – August 2024

Competitor

- Participated in national and international physics competitions, won several awards and medals.
- Represented Turkey in the national teams for 54th International Physics Olympiad, 8th European Physics Olympiad, 23rd and 24th Asian Physics Olympiads.
- Awarded monthly scholarship for undergraduate studies by the Turkish Scientific and Technological Research Organization for achievements in representing Turkey in international competitions.

2024 İstanbul Physics Olympiad Experiment Exam

March 2024

Lead Organizer

- Tested experiment sets, wrote instructions, hints and solutions for the problems.
- Organized and conducted the six-hour-long experimental exam.
- Gave a presentation on the problem solutions in a teacher training program.

SciTech Olympiad Society

Sept 2019 – August 2024

Volunteer

- Volunteered at the student led organization aimed to make olympiad physics accesible.
- Tutored high school students for olympiad level physics and mathematics.
- Translated physics education materials to Turkish.
- Wrote problem sets and solution manuals for national and international physics olympiad problems.

Awards & Honors

- Excellence Scholarsip (2025-2026, École Polytechnique)
- Undergraduate Education Scholarship (2024-2027, TÜBİTAK 2205)
- Silver Medal in International Physics Olympiad (2024, Iran)
- Gold Medal in European Physics Olympiad (2024, Georgia)
- Silver Medal in Asian Physics Olympiad (2024, Malaysia)
- Honorable Mention Award in Asian Physics Olympiad (2023, Mongolia)
- Silver (2023) and Bronze (2022) Medals in Turkish National Physics Olympiad
- 3rd Place in TUBITAK High School Climate Research Projects Competition 2022 (Agriculture Field)
- 5th Team in Fyziklani 2024 Physics Competition
- 3rd Place in NASA Space Apps Hackathon (Regional)
- Bronze Medal in Physics Cup 2024 ([Best Solution Award](#))
- 4th Team in Category C of Physics Brawl Online Physics Competition 2021

Technical Skills

- Python
- MATLAB
- ROOT

Activities

- Bachelor Music Committee (In charge of organizing events, concerts, managing soundchecks.)
- Drums (10 years experience, involved in many short term bands.)
- Rowing (School racing team, FFSU Licensed)
- Bouldering (Indoor Rock Climbing)

Languages

- English (TOEFL iBT: 118/120)
- French (B2, TCF: 410/699)
- Turkish (Native)
- German (Beginner)